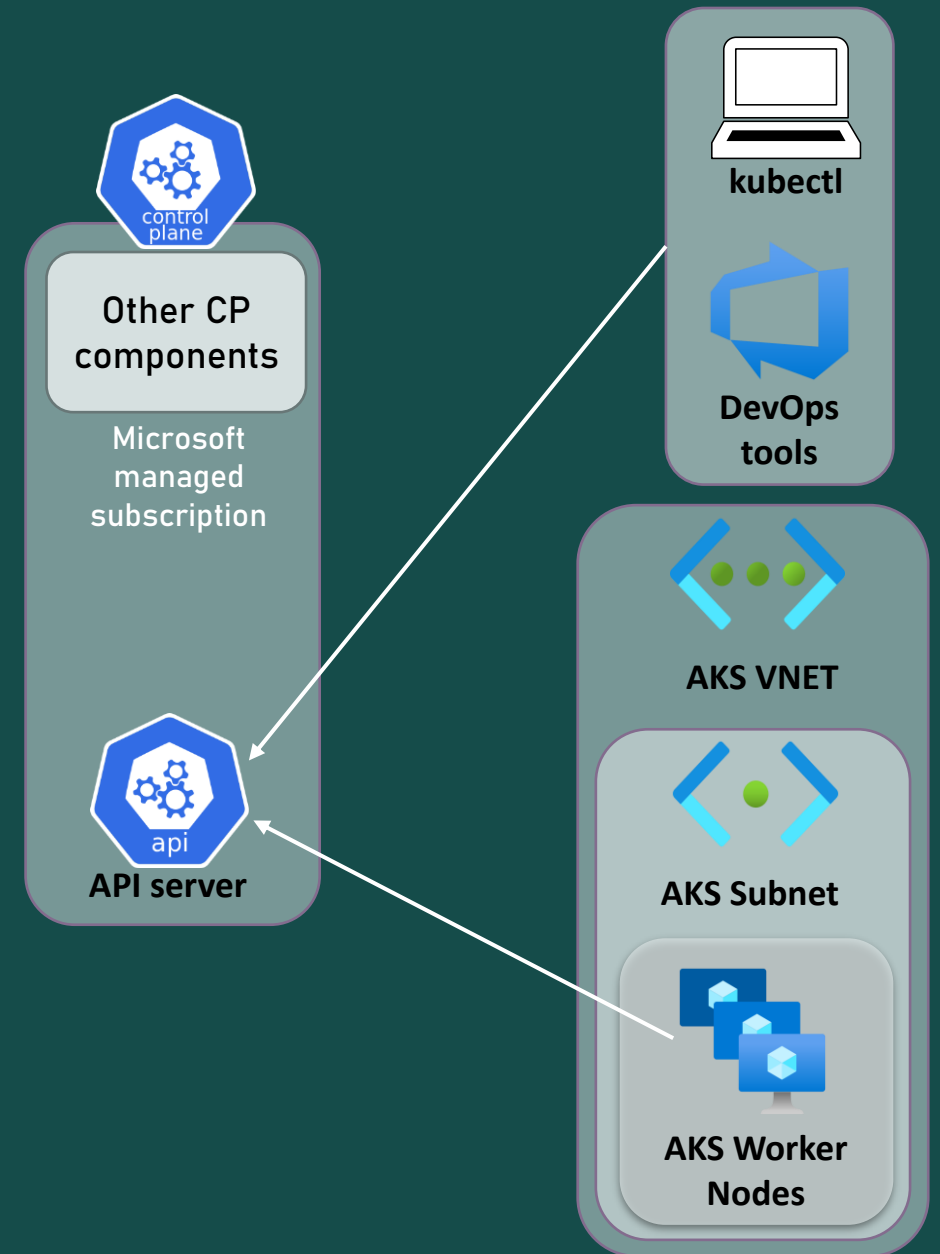


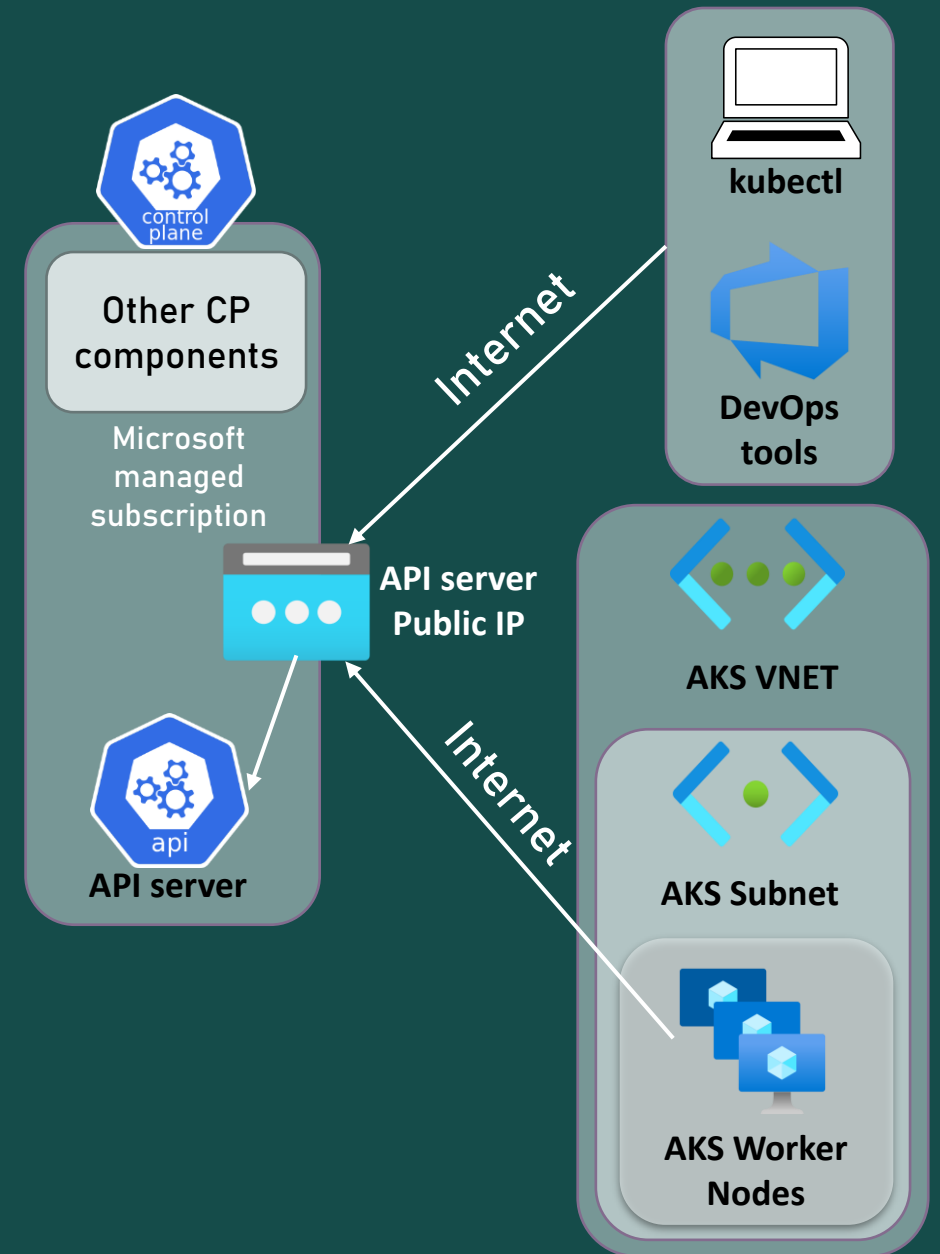
Who and what needs to access the API server?

- kubectl
- DevOps tools
- Components inside AKS worker nodes, for example kubelet



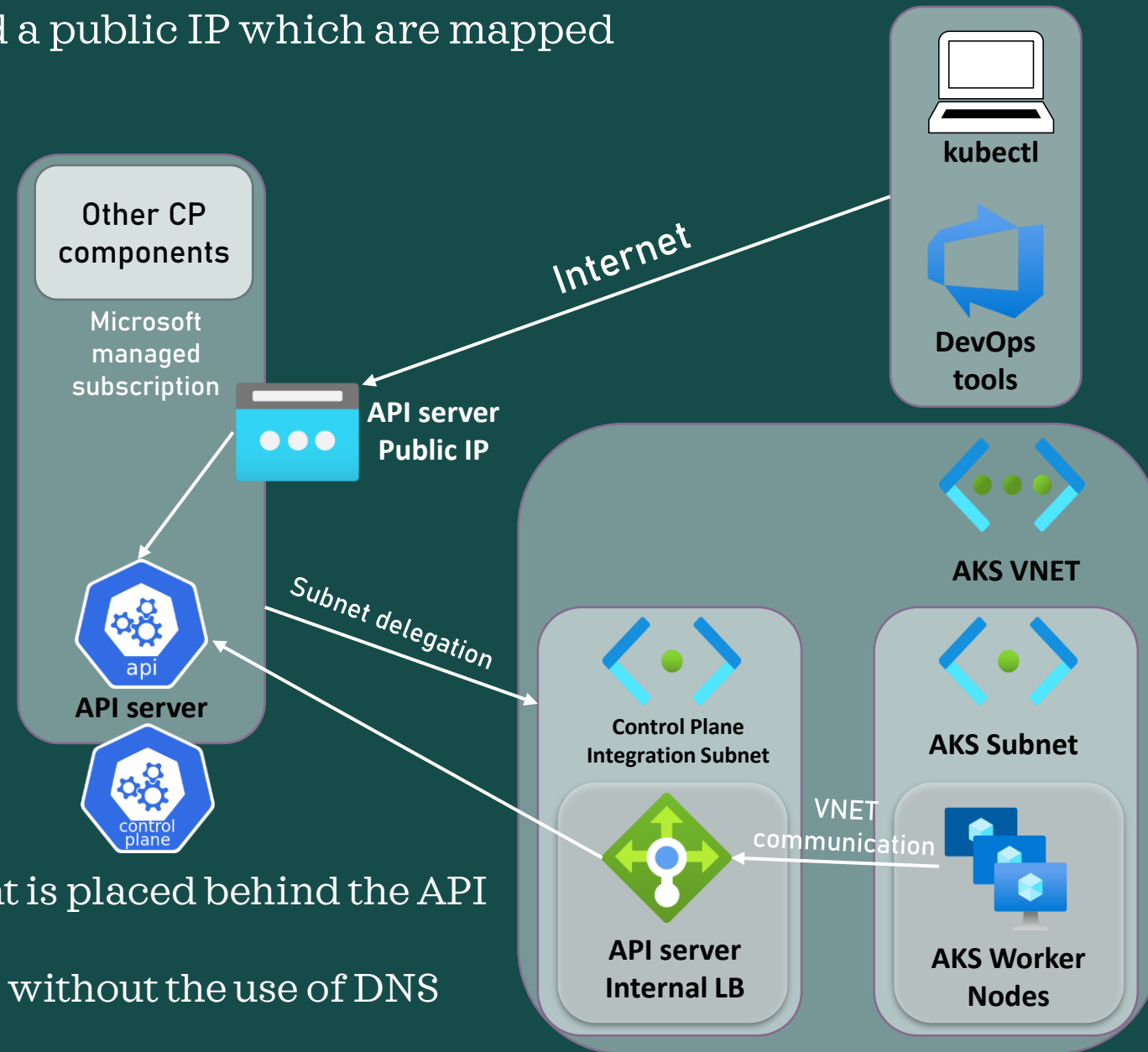
Public AKS cluster

- Public AKS cluster has a public FQDN and a public IP which are mapped
 - The public IP can change
 - The FQDN will not change
- Everyone can connect to the API server if (s)he has the kubeconfig
- We and our DevOps tools connect via public Internet
- Worker nodes connect to the API server via public Internet



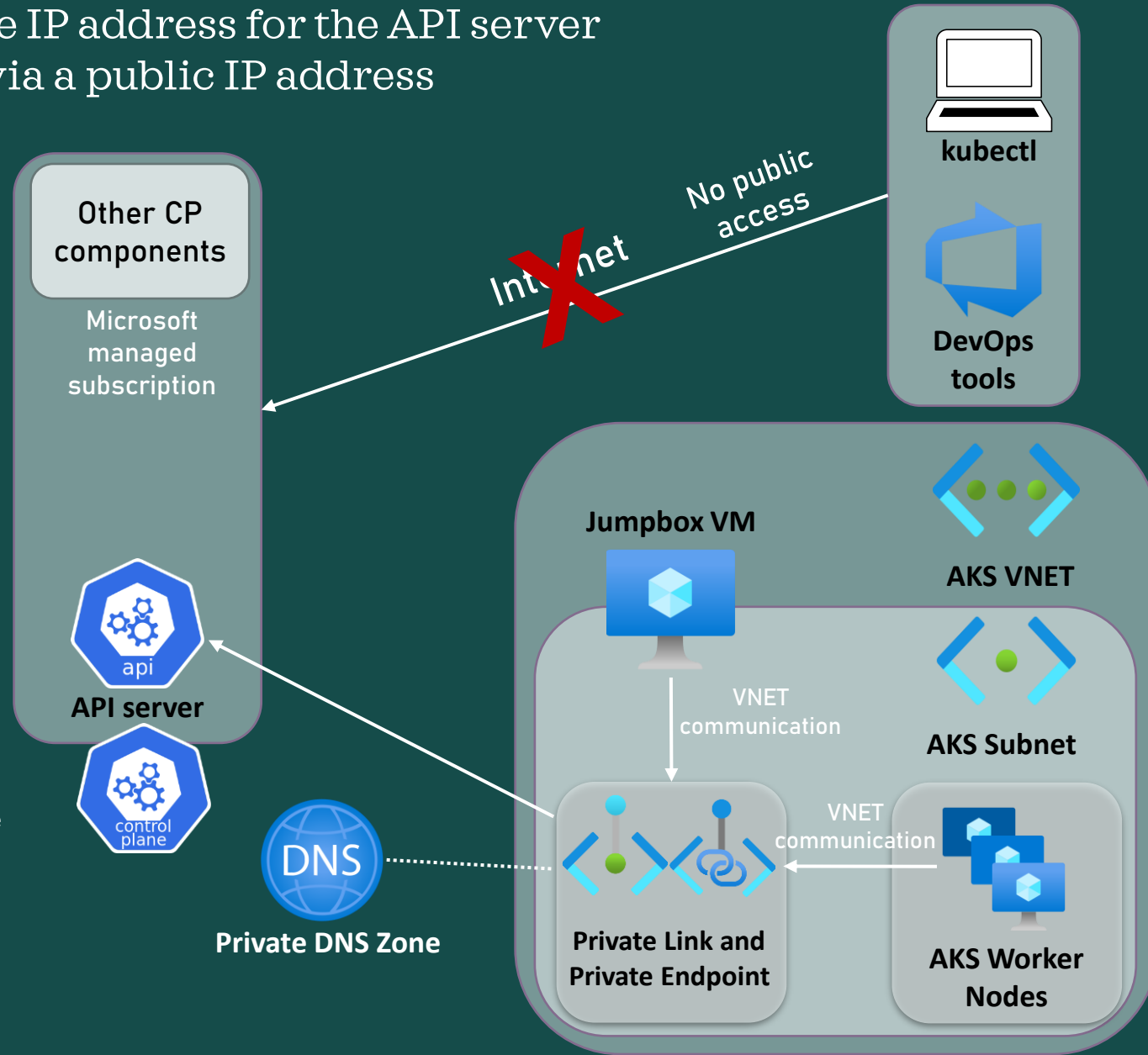
Public AKS cluster with API Server VNET integration

- Public AKS cluster has a public FQDN and a public IP which are mapped
 - The public IP can change
 - The FQDN will not change
- Everyone can connect to the API server if (s)he has the kubeconfig
- We and our DevOps tools connect via public Internet
- The access to the API server can be restricted to specific ranges.
- Worker nodes connect to the API server privately
 - The API server is projected into a delegated subnet in AKS's VNET
 - They use and Internal Load Balancer that is placed behind the API server in the delegated subnet
 - They connect directly to ILB's private IP, without the use of DNS



Private AKS cluster

- Private AKS cluster = no public, but private IP address for the API server
 - Worker nodes will still go to Internet via a public IP address
- A public FQDN can exist or be disabled
- Worker nodes connect to the API server privately using a Private Link and a Private Endpoint exposed in AKS's subnet
- A private FQDN exists with the help of a private DNS zone
- If using custom DNS server:
 - Add 168.63.129.16 as the upstream DNS server
 - Virtual network link for custom DNS server's VNET in the private DNS zone
- Only devices with connectivity to the AKS VNET can connect to the API server



Private AKS cluster with API Server VNET integration

- The same as for normal Private AKS cluster:
 - A public FQDN can exist or be disabled
 - Only devices with connectivity to the AKS VNET can connect to the API server
 - A private FQDN exists with the help of a private DNS zone
 - Custom DNS server needs 168.63.129.16 and VNET link in the private DNS zone for its VNET
- The same as for public AKS cluster with API server VNET integration:
 - Worker nodes connect to the API server privately using the API server Internal Load Balancer

